

AI In Recruitment

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AI usage in organizations has seen significant growth over the last three years, rising from 55% of organizations using AI in at least one business function in 2023 to 88% in 2025 (see Figure 1). This corporate adoption is mirrored by a shift in the labour market, where demand for AI skills in job postings has grown significantly across nearly every country, with Canadian postings alone increasing 4.2 times within a single year (see Figure 2). It's undeniable that AI is growing to be a more significant part of organizational function. While much of its usage in business remains experimental, companies are clearly eager to reap the efficiency gains and cost savings that AI promises. One of the primary areas for these experiments is Human Resources. A 2024 survey found that among companies experimenting with AI or GenAI, 70% were doing so in HR, with talent acquisition ranking as the top use case (Boston Consulting Group, 2025). AI recruitment software has also seen a significant spike, with estimates that market size will grow to over \$1 billion by 2030 (see Figure 3). Yet as excitement around AI continues to grow, so does public unease. On average, 34% of adults say they are more concerned than excited about the increased use of AI, while an additional 42% are equally concerned and excited (Poushter, Fagan, & Corichi, 2025). The unease is difficult to dismiss – especially when AI is making decisions about people's livelihoods.

AI Sentiment in Recruitment

Public Sentiment on AI remains largely divided. Many organizations have shifted business goals to be centered around AI, with 50% of companies on major stock exchanges mentioning AI in their earnings calls in October 2025 (see Figure 4). This enthusiasm is not shared by the general public – an NBC poll conducted in March 2026 revealed that 57% of respondents believe the risks of AI outweighs the benefits, with 46% saying they hold negative

feelings towards the concept of AI (Smith, 2026). This tension is particularly visible in the recruitment sector. Although an estimated 88% of organizations utilize AI for candidate screening (Amitabh & Ansari, 2025), 80% of managers report a dislike for AI-generated CVs and job applications (Robinson, 2024). Candidates express a similar negativity: 45% oppose AI-driven application reviews, and 71% are firmly against the use of AI for final hiring decisions (Rainie, Anderson, McClain, Vogels, & Gelles-Watnick, 2023). Ultimately, this disconnect suggests that while AI adoption is growing rapidly, a significant trust gap exists between organizations deploying these tools and the people affected by them.

The Benefits and Risks of AI in Recruitment

Despite public negativity regarding AI, its prominent role in Human Resources is primarily driven by its ability to perform administrative tasks with greater speed and efficiency. Recent estimates suggest that such automation can reduce time-to-hire by over 75% (Ideatory Pte, 2025) – a critical advantage for organizations competing for the best talent. By streamlining these workflows, HR professionals can pivot from administrative and repetitive tasks to higher-value strategic initiatives. Furthermore, AI systems can process applications at a significantly higher volume than the capacity of manual teams. Beyond efficiency, platforms like SeekOut and hireEZ claim to enhance talent sourcing by reaching broader, more diverse candidate pools, increasing the likelihood for firms to find the right fit for their roles. Perhaps the most significant – and debated – benefit is the objectivity of AI. Research by the Harvard Business Review found that algorithm-based decisions were 25% more accurate at predicting job performance than human judgement (Kuncel, Ones, & Klieger, 2014). Key supporters of these systems argue that AI has the ability to reduce bias by making decisions based on job-relevant information rather than personal feelings and judgements.

On the other hand, deeper adoption of agentic AI systems is met with some hesitancy. The Human Resources Professionals Association (2025) has highlighted significant risks inherent in these tools, specifically the “black box” nature of AI and its lack of explainability. In many cases, these systems provide outputs without substantial information regarding the evaluation methods used or the logic behind a specific decision. Furthermore, data privacy remains a pervasive concern that extends beyond the HR industry. Currently, only an estimated 37% of consumers trust AI with their personal data, and 70% of adults say they do not trust companies to utilize the technology responsibly (Protecto, 2025). This lack of trust threatens adoption, and companies that fail to demonstrate digital responsibility risk losing high-quality candidates who may decline interviews with them. Governments and regulatory bodies have already begun responding to these concerns. For instance, the Government of Ontario now requires employers to disclose the use of artificial intelligence to their applicants, providing a layer of transparency that was previously missing from the process. (Working for Workers Four Act, 2024). However, transparency does not resolve the underlying issue of algorithmic integrity. Multiple studies have expressed concern regarding the risk of bias and discrimination embedded within training data. Because models are trained on historical, real-life data, they risk perpetuating existing systemic biases and inadvertently promoting unequal opportunities for applicants (Rigotti & Fosch-Villaronga, 2024). Researchers are divided on whether AI systems serve as objective judges or as a magnifying glass for human prejudice. While algorithms can eliminate the risk of inconsistencies in human recruiters, they may also code historical exclusions into decision systems. This raises a critical question: to what extent can AI recruitment systems make fair and accurate hiring decisions given the limitations of real-world data?

AI Discrimination in Practice

One way these limitations are showing itself is in the active litigation: Mobley v. Workday. In February 2023, the plaintiff Derek Mobley filed suit against Workday, alleging that it's AI screening tool "discriminated against him [and others similarly situated] based on race, disability, and age" (Wagner, 2025). Affected applicants reported having their applications rejected within minutes of their submission, and often late at night. This is one of the first legal cases of AI discrimination in hiring, and its outcome is expected to set an important precedent for the future use of AI in recruiting. In response to these claims, a spokesperson for Workday denied these allegations, stating: "This lawsuit is without merit. ...Our AI capabilities look only at the qualifications listed in a candidate's job application and compare them with the qualifications the employer has identified as needed for the job. They are not trained to use—or even identify—protected characteristics like race, age, or disability." (Asare, 2025).

Mobley v. Workday is not the only example of AI discrimination in recruiting tools. In 2015, Amazon discontinued an experimental AI tool it had developed that rated job applicants, in hopes to streamline their identification of top talent. The model was trained on resumes historically submitted to the company, which were predominantly from men - a reflection of the longstanding gender imbalance in the tech industry. As a result, the model learned to favour male candidates, penalizing resumes where gender could be inferred through references to women's clubs or all-female colleges (Dastin, 2018).

These cases illustrate a concept known as proxy discrimination. Even when protected characteristics are removed, AI systems can infer them through correlated variables. For example, work history as a proxy for age, postal code as a proxy for race, or listed hobbies as a proxy for gender. This is particularly concerning because proxy discrimination often "disproportionately harms members of a protected class." (Prince & Schwarcz, 2020),

demonstrating that AI systems do not need to be explicitly trained to misuse data in order to do so.

Experiment: Measuring Gender Bias in Job Title Embeddings

These concerns on the persistence of discrimination through proxies raises an important question: are there existing associations and biases already embedded into the models underlying AI recruitment tools? The following experiment measures the gender associations of job titles in a language embedding model, to evaluate if patterns of bias or historical injustices exist.

Methodology

To prevent the risk of model-specific associations, the experiment was conducted across two separate embedding models: Google's Gemini Embedding 001 and OpenAI's Text Embedding 3 Small. Results were consolidated by averaging the normalized scores for each job title across both models. A diverse set of 100 job titles was generated using Gemini, spanning a range of industries and seniority levels. Only job titles themselves were embedded, excluding industry category labels. Gender was represented using two terms: "Female" and "Male".

Initial associations were measured using cosine similarity. However, raw cosine scores were insufficient, as some terms sit closer to all job titles in the embedding space. To control for this, scores were converted to z-scores, measuring each job title's gender association *relative to the average* across the full dataset. A final association score was calculated for each job title by taking the difference between its male and female z-scores, where a positive value indicates a male association and a negative value indicates a female association (see Figure 6).

Implementation

The main program was implemented in Typescript, and deeper analysis was conducted in Microsoft Excel. Job titles were store in a JSON file and loaded into a string array, which was

then passed to both the Gemini Embedding 001 and OpenAI Text Embedding 3 Small models to generate vector representations. The terms “Female” and “Male” were embedded using the same process, and cosine similarity scores between each job title and each term were saved into a CSV file. In Excel, cosine similarity scores from both models were averaged for each job title before normalization. Industry categories were mapped to each job title using VLOOKUP, and the mean and standard deviation were calculated across all jobs for Female and Male separately. Z-scores were calculated for each job and gender term, and a final association score was computed by taking the difference between each job title’s male and female z-scores. This information was used to generate bar charts for visual display of the results (see Figure 6).

Results

Of 100 job titles analyzed, 49 leaned towards “Female” and 51 leaned towards “Male”. The job title that leaned closest to Female was Medical Receptionist with a score of -1.87. On the other hand, Judge leaned closest to Male with a score of 1.64. In examining industries, all job titles under Administration and Human Resources leaned closer to Female, while all job titles under Trades and Management leaned closer to Male. Two job titles had a score of 0: Product Manager and Real Estate Agent.

Discussion and Analysis

The results of this experiment reflect some of the historical imbalances of representation across industries rather than the current or ideal state of the workforce. Management was a particularly notable area, with all management-level job titles associating more closely with “Male” (see Figure 6c). This is consistent with real-world data: only approximately 6.04% of companies on the S&P 500 had a female CEO in 2020 (Bell & Llewellyn, 2021), and only 30.8% of women held senior management positions as of 2021 (Gueye, 2024). The model has not

learned what the workforce should look like – it has learned what it has historically looked like.

The convergence of this pattern across two separate models suggest that this is not an outcome of any single model, but something deeper in the data that these models are trained on.

While these patterns reflect historical structures that are not inherently ‘incorrect’, they carry meaningful implications for AI recruitment systems. Biases can enter the pipeline at the foundational model level, before any hiring-specific logic is applied – meaning even well-intentioned systems are not immune. In the case of Workday, even if their model is not explicitly trained to identify protected characteristics, their systems could still reproduce discriminatory outcomes regardless. And while these biases could be unintentional, it doesn’t matter if the effect on candidates is the same.

This experiment has several limitations worth acknowledging. First, an association made does not prove hiring bias – it illustrates what embedding models have learned from language patterns. Additionally, two models represent a limited sample size, and does not suggest that all embedding models would be reflect the same results. The job titles themselves were generated using an LLM, which could potentially have introduced its own selection bias in the titles included. These limitations suggest that the findings are more illustrative rather than conclusive of bias. However, the consistency of the patterns observed does point to a structural problem that warrants our attention. If bias is present at the model level, addressing it requires more than removing protected characteristics from inputs – it requires transparency in how models were selected, evaluated and audited throughout the recruitment pipeline.

Designing for Transparency

Many research studies have explored frameworks and methods to increase transparency in AI systems. Explainable AI (XAI) is the name of this research area that addresses “the black-

box nature of complex AI models, particularly as these systems increasingly influence high-stakes domains such as healthcare, finance, and autonomous systems.” (Chinnaraju, 2025). Work in this field is dedicated to understanding the internal mechanisms of a model – and being able to interpret why a decision system produced a particular output. While still an emerging field, XAI could help professionals in the recruiting space better understand how automated decisions are made, and how to design their systems and hiring pipelines accordingly.

NLP evaluation is another critical research area focused on designing reliable evaluation processes for AI outputs. Where it differs from XAI is in its approach: rather than examining a model’s internals, it implements structure on the evaluation process, so that regardless of which model is used, the process remains consistent and auditable. One notable method in this research space is Autorubric (Delip & Chris, 2026), an open-source Python framework that defines an analytical rubric for AI evaluation. It breaks evaluation down into separate, independently sourced criteria, and differs from other methods by moving beyond theory into a directly implementable system.

The Autorubric methodology follows a few key steps. First, each criterion must fit one of three types: binary (MET/UNMET), ordinal (a defined scale e.g. Poor/Fair/Good/Excellent) or nominal (categorical labels). Second, criteria must be operationally precise. Rather than relying on vague definitions that are open to interpretation, criteria are framed concretely (e.g. the candidate's written response uses specific examples to illustrate past experience). Third, negative criteria is included to counteract leniency bias. For example, “The candidate makes vague, unsupported claims such as ‘I am a great team player’” would trigger a point deduction if MET. Fourth, each criterion is evaluated independently, in a separate AI call, reducing the risk that strength in one category incorrectly inflates the scores in other categories. Fifth, a multi-judge

approach is used, where multiple AI models make evaluate each criteria, and a majority vote determines the final verdict. Sixth and finally, is that each judge is to required to provide a written rationale for its score, creating an audit trail that human reviewers can inspect.

Both XAI and NLP overlap in their underlying goals: transparency and accountability in automated decision-making. Together, they can be used as complementary tools. XAI helps us understand and question what our model is doing internally, while structured evaluation frameworks help us govern what the model does in practice. In the context of recruitment, organizations might use analytical rubrics to enforce fair screening criteria while simultaneously referring to XAI research to better understand the biases embedded in the models they use. Regardless of the methods used, neither approach eliminates the need for human oversight.

Conclusion

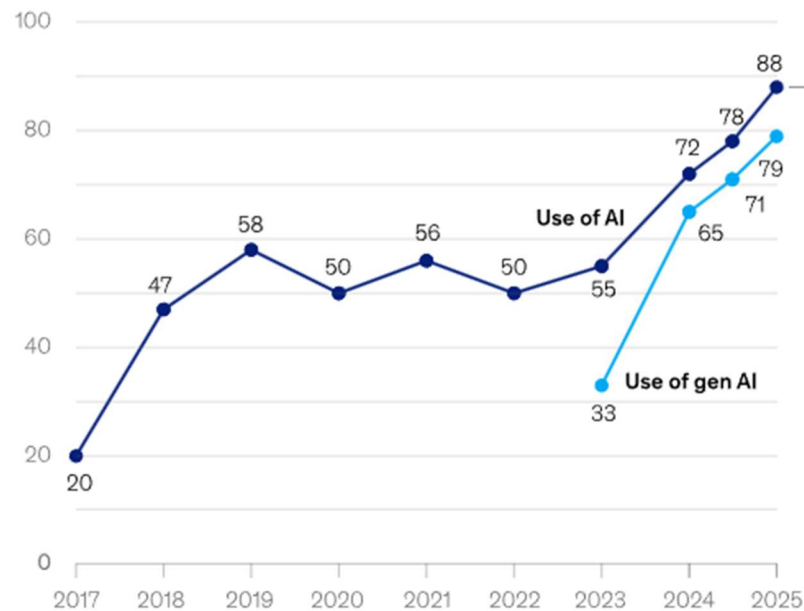
The question of whether AI recruiting systems can make fair and accurate hiring decisions does not have a simple answer. Real-world hiring data carries historical biases and inequality, that without deliberate intervention, could be reproduced – and even amplified. While frameworks and methods exist to increase the transparency and accountability of these systems, these solutions alone are not sufficient. Organizations must also keep in mind the human experience of being evaluated by an algorithm. Given the distrust of AI driven systems, companies have a responsibility to be thoughtful in the way they use and disclose the use of these tools. As cases like *Mobley v. Workday* illustrate, the legal and ethical responsibility for hiring decisions ultimately rests with people, and any AI system used in that process must be designed with meaningful human oversight at every stage. The future of AI recruitment is to learn to develop it responsibly. Until these systems can consistently demonstrate equitable and fair outcomes, they must be used as tools *to support human judgement*, not replace it.

Figure 1

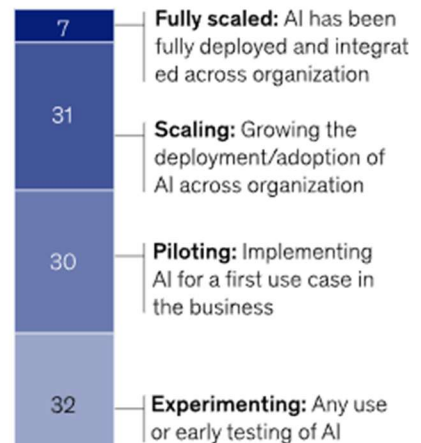
Reported use of AI in at least one business function continues to increase.

Use of AI by respondents' organizations, % of respondents

Organizations that use AI in at least 1 business function¹



Phase of AI use among organizations using AI in 2025



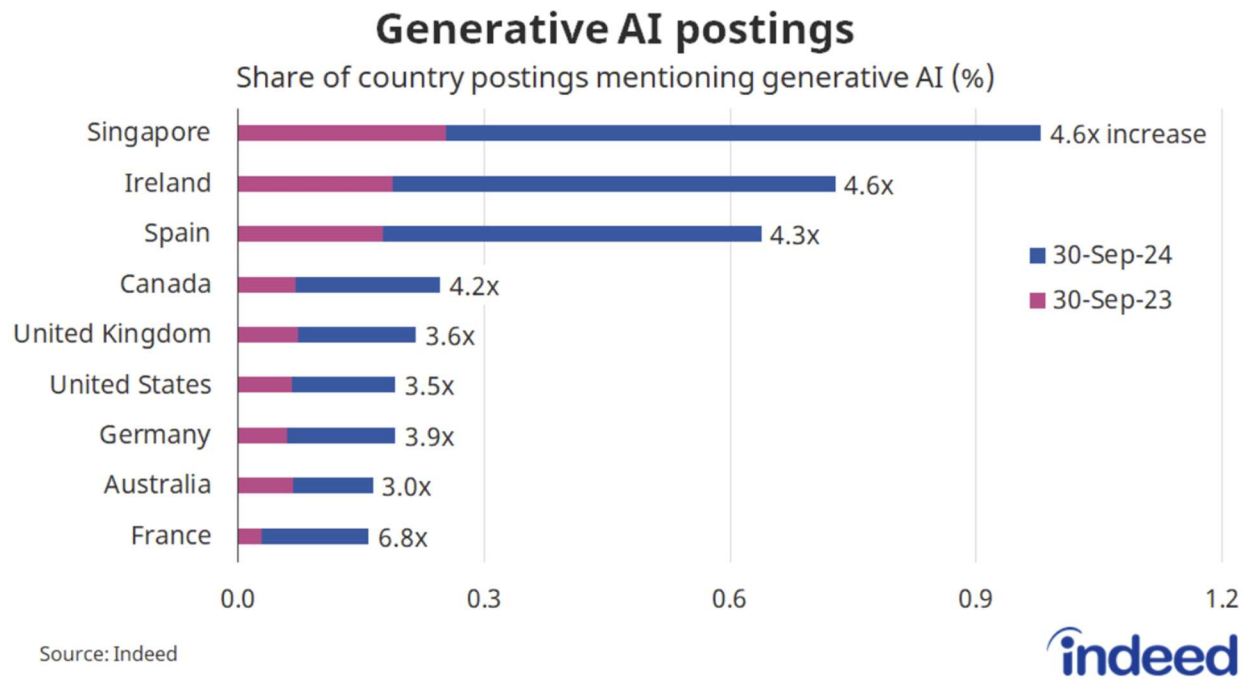
¹In 2017, the definition for AI use was using AI in a core part of the organization's business or at scale. In 2018–19, the definition was embedding at least 1 AI capability in business processes or products. From 2020, the definition was that the organization has adopted AI in at least 1 function, and in 2025, the definition was regular use of AI in at least 1 function.

Source: McKinsey Global Surveys on the state of AI, 2017–25

McKinsey & Company

Note. From McKinsey & Company. (2025). The state of AI in 2025: Agents, innovation, and transformation. McKinsey & Company. Retrieved from <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai/>

Figure 2

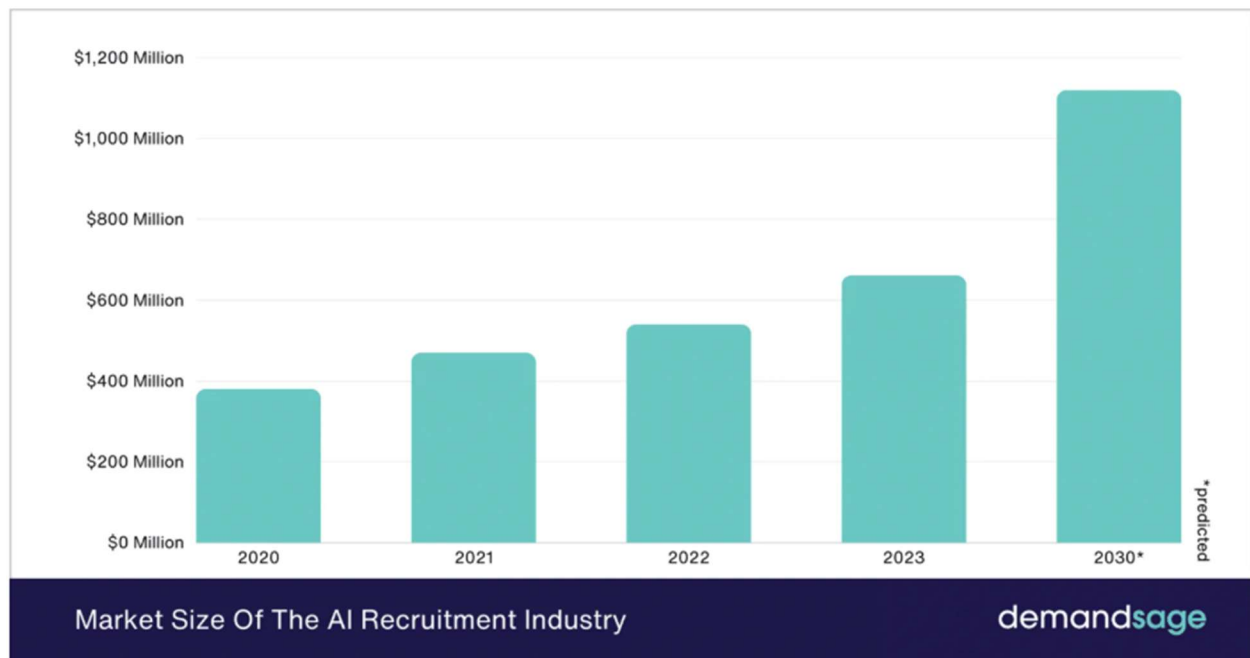


Note. From Pickering, C. (2024, December 5). Rapid Growth in GenAI Job Postings:

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<https://www.hiringlab.org/au/blog/2024/12/05/rapid-growth-in-genai-job-postings-expectations-and-surprising-trends/>

Figure 3

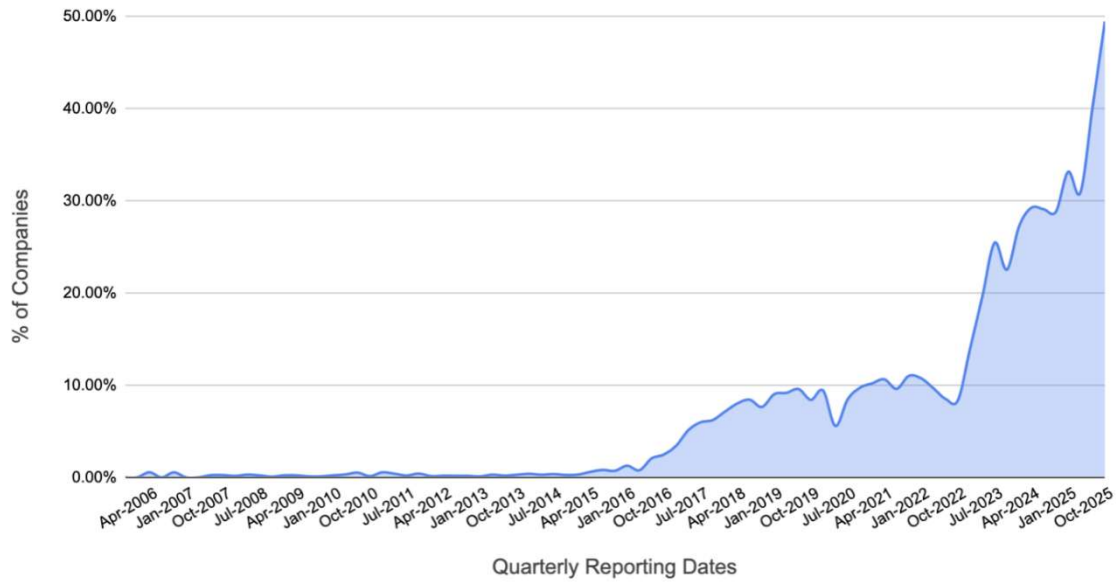


Note. From Kumar, N. (2026, January 8). AI Recruitment Statistics 2026 (Global Data & Trends). Retrieved from DemandSage: <https://www.demandsage.com/ai-recruitment-statistics/>

Figure 4

Percentage of Companies Mentioning AI in Earning Calls

Quarterly earning transcripts mentioning AI at least once. Source: Earning Call Transcripts on Snowflake Marketplace



Note. From Fruit Stand. (n.d.). Earnings Call Transcripts. Snowflake Marketplace. Retrieved from <https://app.snowflake.com/marketplace/listing/GZTYZ40XYU5/fruit-stand-earnings-call-transcripts>, as visualized by fruitsanddev. (2025, November 30). 50% of Companies mentioned AI at least once in their earning calls so far this quarter. (r/dataisbeautiful). Retrieved from https://www.reddit.com/r/dataisbeautiful/comments/1payrgs/oc_50_of_companies_mentioned_ai_at_least_once_in/

Figure 5

Methodology on Semantic Search

$$z_{score} = \frac{\text{cosine similarity (one career)} - \text{mean (of all careers)}}{\text{Standard Deviation}}$$

Z-score Difference: $\Delta z(c) = z_M(c) - z_F(c)$

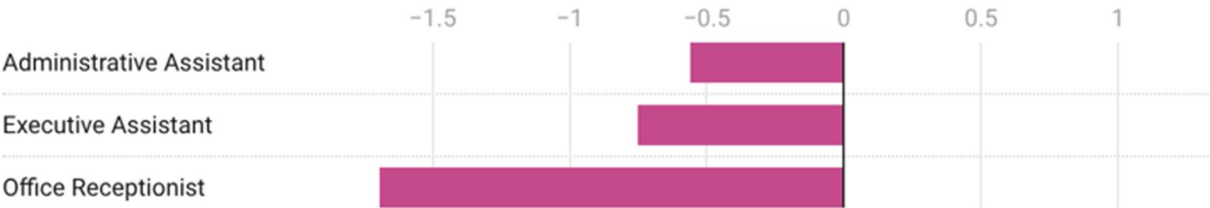
$$\Delta z(c) < 0 = \text{Closer to Female}$$

$$\Delta z(c) > 0 = \text{Closer to Male}$$

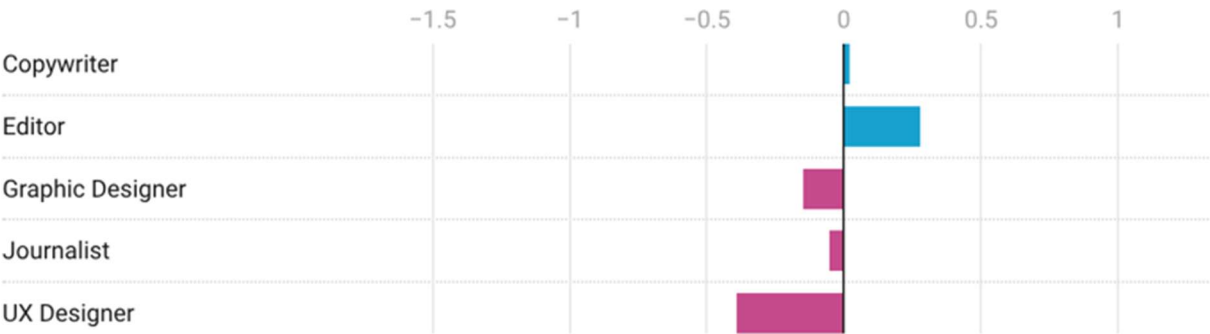
Figure 6a

Semantic Mapping Between Job Titles and Gender

Administration



Creative



Education

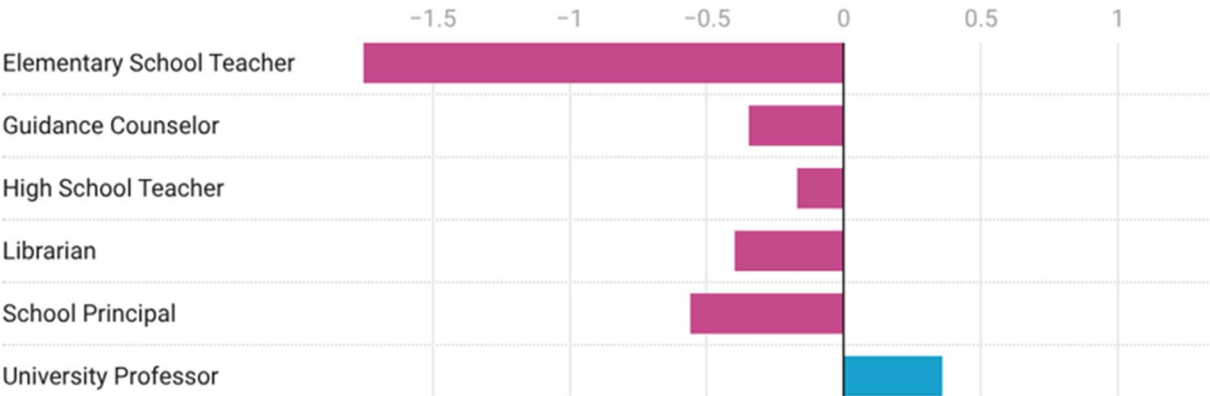


Figure 6b

Healthcare

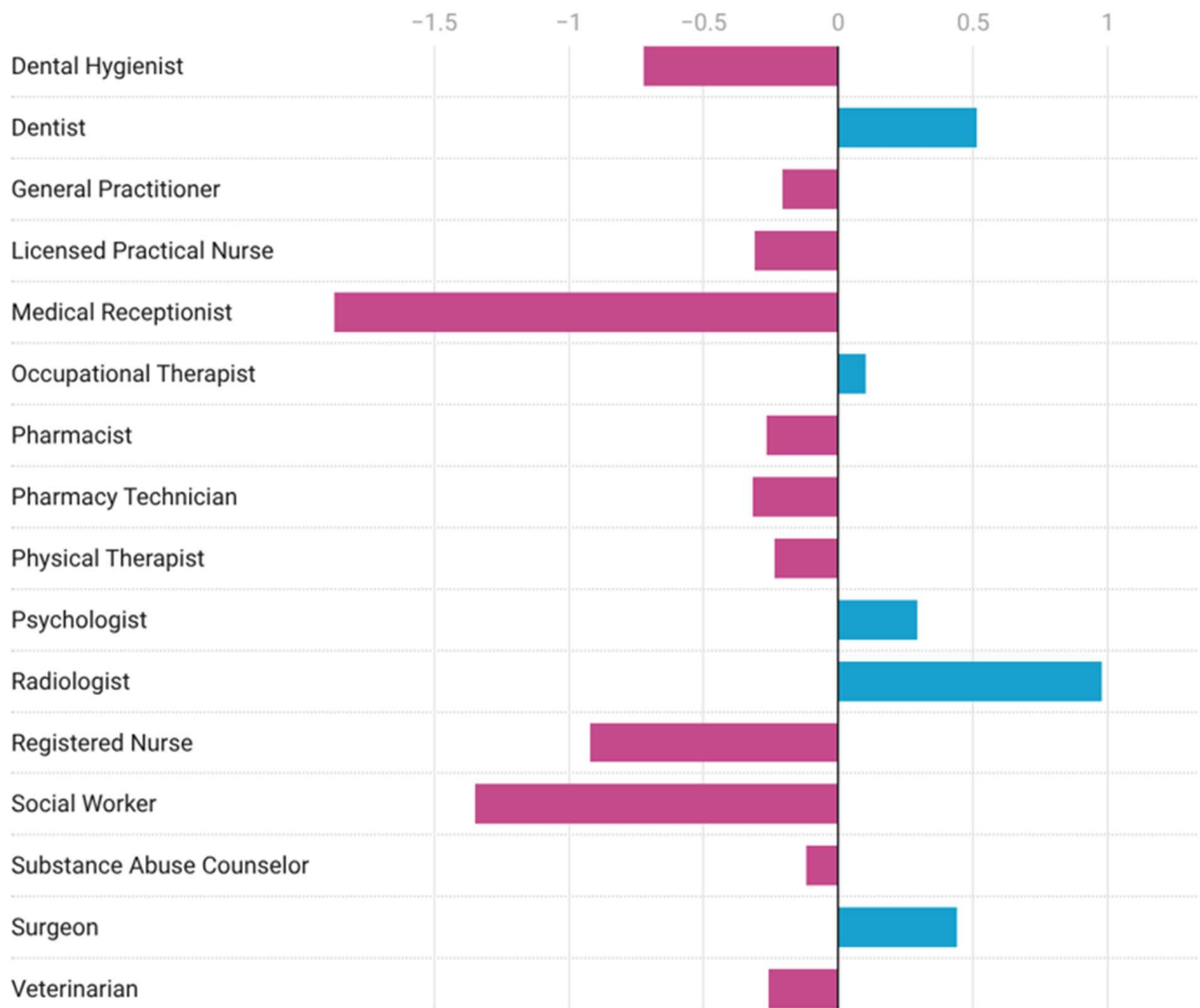


Figure 6c

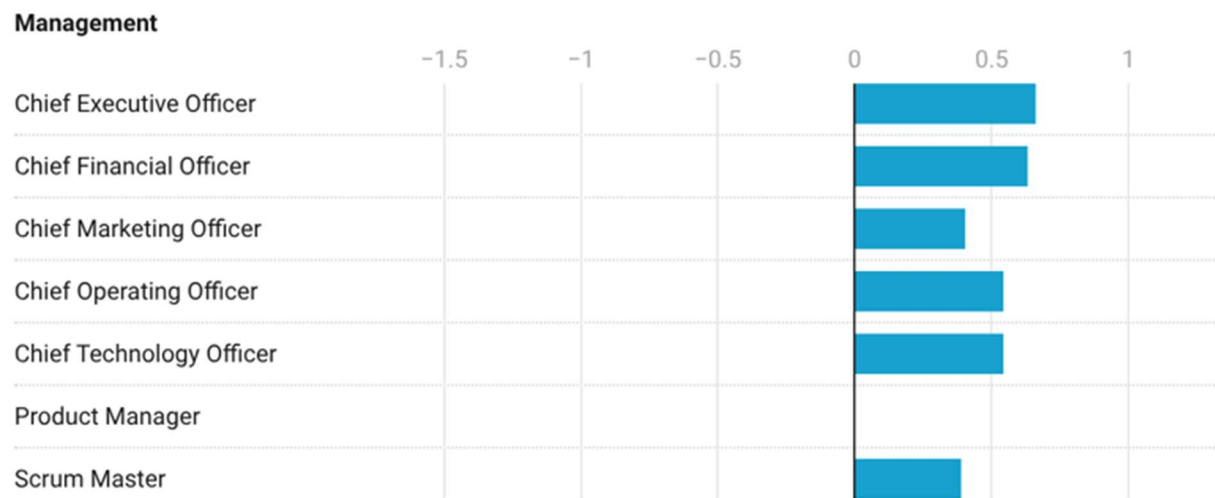


Figure 6d

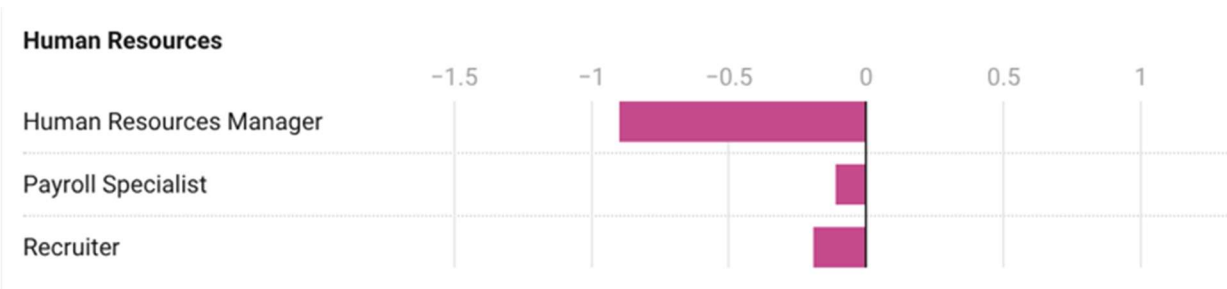


Figure 6e

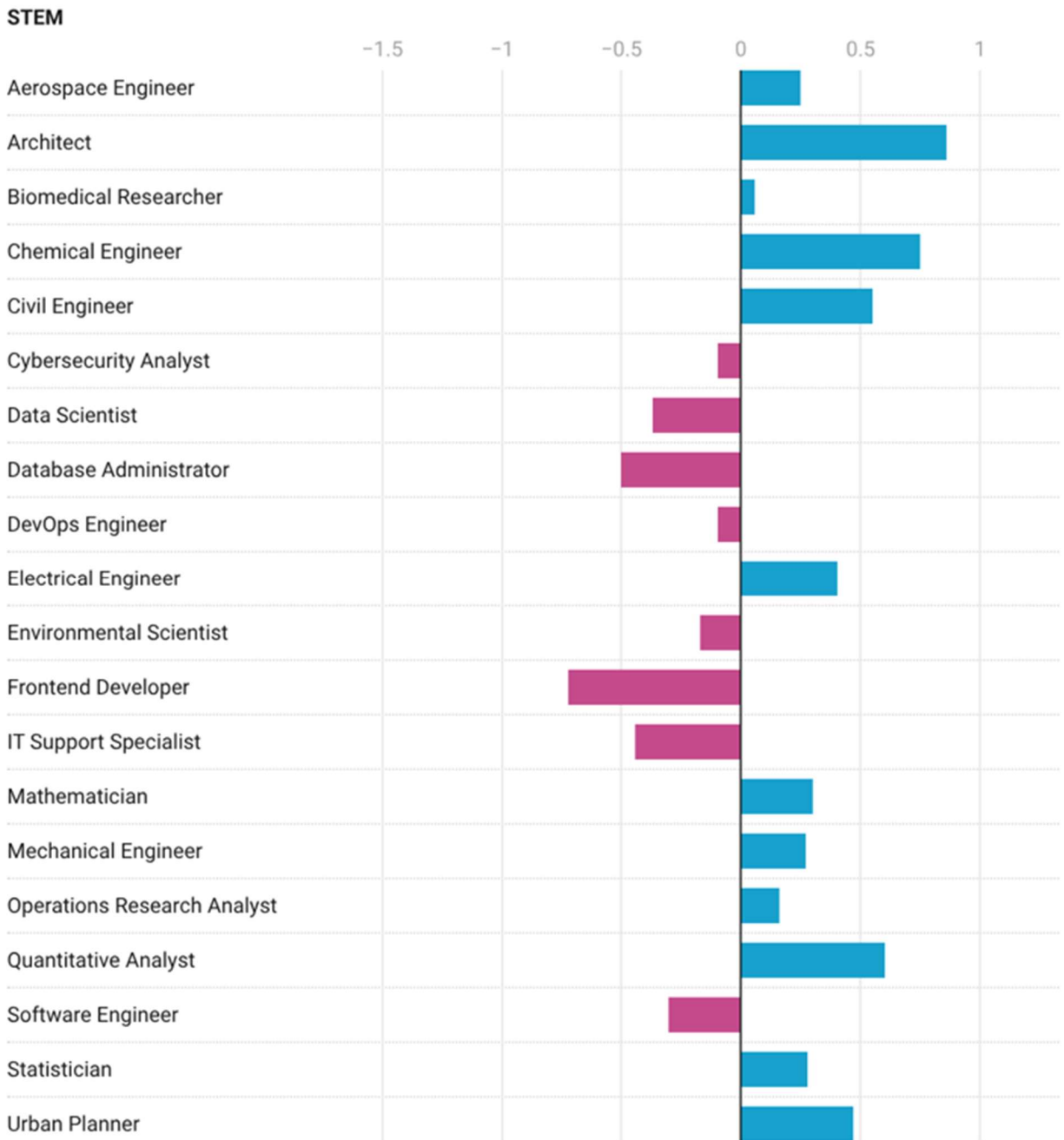


Figure 6f



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